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I pledge my honor that I have abided by the Stevens Honor System. Octavio Morales

4. a. The function calculates the sum of squares from 1 to n ($\sum_{k=1}^{n} k^2$).

   b. Its basic operation is multiplication

   c. The basic operation is performed n times.

   d. The efficiency class is Θ(n).

   e. Due to this (the image below), the summation can be represented by $\frac{n(n+1)(2n+1)}{6}$,

which has an efficiency class of Θ(1).

4. e. $x(n) = x(n-1) + n^2$ for $n > 0$ $x(0) = 0$

S. 1. $x(n-1) = x(n-2) + (n-1)^2$

$x(n) = x(n-2) + (n-1)^2 + n^2$

S. 2. $x(n-2) = x(n-3) + (n-2)^2$

$x(n) = x(n-3) + (n-2)^2 + (n-1)^2 + n^2$

S. 3. $x(n) = x(n-i) + (n-i+1)^2 + (n-i+2)^2 + \dots + (n)^2$

S. 4. $n - i = 0$

$i = n$

$x(0) = 0$

S. 5. $x(n) = 0^2 + 1^2 + 2^2 + 3^2 + \dots + n^2$

$x(n) = \frac{n(n+1)(2n+1)}{6}$

1.a. $x(n)=x(n-1)+5,\ x(1)=0$

S.1. $x(n-1)=x(n-2)+5$
$x(n)=x(n-2)+5+5$

S.2. $x(n-2)=x(n-3)+5$
$x(n)=x(n-3)+5+5+5$

S.3. $x(n)=x(n-i)+5i$

S.4. $n-i=1$
$i=n-1$

S.5. $x(n)=x(n-n+1)+5(n-1)$
$x(n)=x(1)+5(n-1)$
$x(n)=0+5(n-1)$
$x(n)=5(n-1)$

b. $x(n)=3x(n-1),\ x(1)=4$

S.1. $x(n-1)=3x(n-2)$
$x(n)=3(3x(n-1))$

S.2. $x(n-2)=3x(n-3)$
$x(n)=3(3(3x(n-1)))$

S.3. $x(n)=3^i x(n-i)$

S.4. $n-i=1$
$i=n-1$

S.5. $x(n)=3^{n-1}x(n-n+1)$
$x(n)=3^{n-1}x(1)$
$x(n)=3^{n-1}\cdot 4$

c. $x(n)=x(n-1)+n,\ x(0)=0$

S.1. $x(n-1)=x(n-2)+(n-1)$
$x(n)=x(n-2)+(n-1)+n$

S.2. $x(n-2)=x(n-3)+(n-2)$
$x(n)=x(n-3)+(n-2)+(n-1)+n$

S.3. $x(n)=x(n-i)+(n-i+1)+\dots+n$

S.4. $n-i=0$
$i=n$

S.5. $x(n)=x(n-n)+(\overset{0}{n-n}+1)+\dots+n$
$x(n)=x(0)+1+\dots+n$
$x(n)=\frac{n(n+1)}{2}$

d. $x(n) = x(n/2) + n,\ x(1) = 1,\ n = 2^k$

s. 1. $x(n/2) = x(n/4) + n/2$

$x(n) = x(n/4) + n/2 + n$

s. 2. $x(n/4) = x(n/8) + n/4$

$x(n) = x(n/8) + n/4 + n/2 + n$

s. 3. $x(n) = x\left(\frac{n}{2^i}\right) + \frac{n}{2^{i-1}} + \ldots + \frac{n}{2^0}$

s. 4. $n = 2^i$

$\log_2(n) = i$

s. 5. $x(n) = x\left(\frac{n}{2^{\log_2(n)}}\right) + \frac{n}{2^{\log_2(n)+1}} + \ldots + \frac{n}{2^0}$

$x(n) = x\left(\frac{n}{n}\right) + \frac{2n}{n} + \ldots + n$

$x(n) = x(1) + n + \frac{2n}{n} + \frac{2^2 n}{n} + \ldots \frac{2^k n}{n}$

$x(n) = 2^{k+1} - 1 = 2 \cdot 2^k - 1$

$x(n) = 2n - 1$

e. $x(n) = x(n/3) + 1,\ x(1) = 1,\ n = 3^k$

$\log_3(n) = k$

s. 1. $x(n/3) = x(n/9) + 1$

$x(n) = x(n/9) + 1 + 1$

s. 2. $x(n/9) = x(n/27) + 1$

$x(n) = x(n/27) + 1 + 1 + 1$

s. 3. $x(n) = x\left(\frac{n}{3^i}\right) + i$

s. 4. $n = 3^i$

$\log_3(n) = i$

s. 5. $x(n) = x\left(\frac{n}{3^{\log_3(n)}}\right) + i$

$x(n) = x(1) + k$

$x(n) = 1 + \log_3(n)$

3. $x(n) = x(n-1) + 2$ for $n > 1$, $x(1) = 0$

a. s.1. $x(n-1) = x(n-2) + 2$
$x(n) = x(n-2) + 2 + 2$

s.2. $x(n-2) = x(n-3) + 2$
$x(n) = x(n-3) + 2 + 2 + 2$

s.3. $x(n) = x(n-i) + 2i$

s.4. $n - i = 1$
$i = n - 1$

s.5. $x(n) = x(n-n+1) + 2(n-1)$
$x(n) = x(1) + 2(n-1)$
$x(n) = 0 + 2(n-1)$
$x(n) = 2(n-1)$

b. The recursive function and the algorithm take the same time to do the calculations.